SUBEXTENSIONS OF A RADICAL EXTENSION OF DEGREE E ARE GENERATED BY POWERS OF THE RADICAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The constant-term argument survives repeated roots: each root beta of P satisfies beta^e = a, so beta/zeta-free form beta = zeta alpha with zeta^e = 1, and zeta = prod zeta_i satisfies zeta^e = 1.

Editorial note. The label lemma-subfields-kummer names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Statement hypothesis: L = K(alpha) of degree e with a = alpha^e in K, and every e-th root of unity in L lies in K
- Same chapter: multiplicativity of degrees in towers (lemma-multiplicativity-degrees)

Lemma 0.1. *Let K be a field. Let L/K be a finite extension of degree e which is generated by an element α with $a = \alpha^e \in K$. If every eth root of unity in L is contained in K, then any sub extension $L/L'/K$ is generated by α^d for some $d|e$.*

Proof. Observe that for $d|e$ the subfield $K(\alpha^d)$ has $[K(\alpha^d) : K] = e/d$ and $[L : K(\alpha^d)] = d$. Let $L/L'/K$ be a subextension. Say $d = [L : L']$. If $\alpha^d \in L'$, then we have $L' = K(\alpha^d)$ for degree reasons. Let $P \in L'[x]$ be the minimal polynomial of α over L'. Then P divides $x^e - a$ and P has degree d. Let us write

$$x^e - a = \prod\nolimits_{i=1,\ldots,e}(x - \zeta_i\alpha)$$

in a splitting field of $x^e - a$ over L. The ζ_i are eth roots of unity and after renumbering we have

$$P = \prod\nolimits_{i=1,\ldots,d}(x - \zeta_i\alpha)$$

The constant term of P is equal to

$$c = (\prod\nolimits_{i=1,\ldots,d}\zeta_i)\alpha^d$$

and is in $L' \subset L$. Since $\alpha \in L$ this implies that $\zeta = \prod_{i=1,\ldots,d}\zeta_i$ is in L and hence in K by our assumption. Thus $\alpha^d = \zeta^{-1}c \in L'$ and we conclude. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.